

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424055
Date	29 September 2026
Day	Tuesday

Set 1: Monthly Sales Data

Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

PROGRAM 1: Line Chart of Monthly Sales

Question

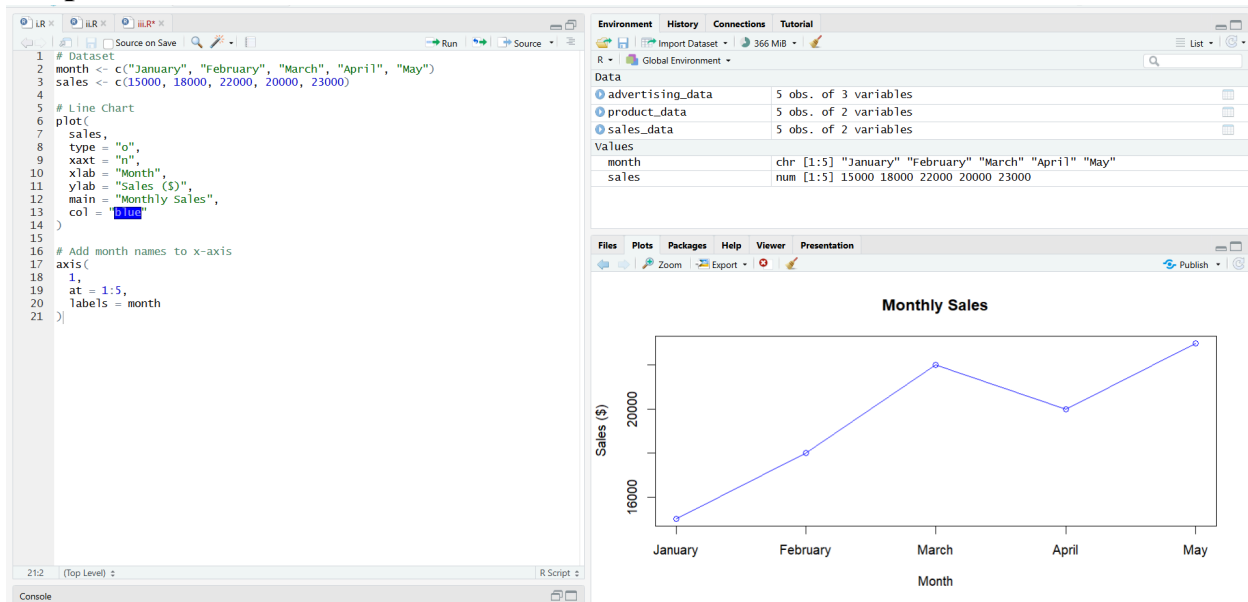
Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

R Program

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Line Chart
plot(
  sales,
  type = "o",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales",
  col = "blue"
)
# Add month names to x-axis
axis(
  1,
  at = 1:5,
  labels = month
)
```

Output



Observation

Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2: Bar Chart of Top-Selling Products

Question

Generate a bar chart to display the top-selling products for the year. Label the chart elements.

Sample Product Data

Product	Annual Sales (\$)
Laptop	45000
Mobile	38000
Tablet	25000
Headphones	18000
Smartwatch	15000

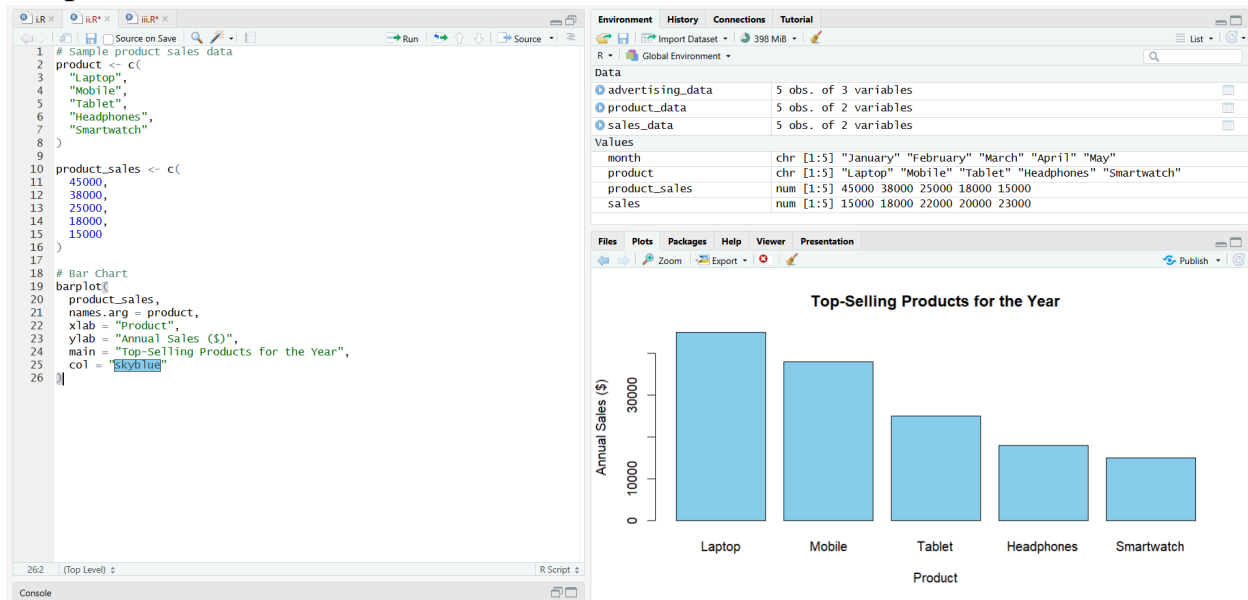
R Program

```
# Sample product sales data
product <- c(
  "Laptop",
  "Mobile",
  "Tablet",
  "Headphones",
  "Smartwatch"
)

product_sales <- c(
  45000,
  38000,
  25000,
  18000,
  15000
)

# Bar Chart
barplot(
  product_sales,
  names.arg = product,
  xlab = "Product",
  ylab = "Annual Sales ($)",
  main = "Top-Selling Products for the Year",
  col = "skyblue"
)
```

Output



Observation

The bar chart shows the sales comparison among the products. Among the sample data, Laptop has the highest annual sales, while Smartwatch has the lowest.

PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

Question

Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

Sample Advertising Data

Month	Advertising Budget (\$)	Sales (\$)
January	5000	15000
February	6000	18000
March	7000	22000
April	8000	20000
May	9000	23000

R Program

```
# Monthly sales data
sales <- c(
  15000,
  18000,
  22000,
  20000,
  23000
)
# Sample advertising budget data
advertising <- c(
  5000,
  6000,
  7000,
  8000,
  9000
)
# Scatter Plot
plot(
  advertising,
  sales,
  main = "Advertising Budget vs Monthly Sales",
  xlab = "Advertising Budget ($)",
```

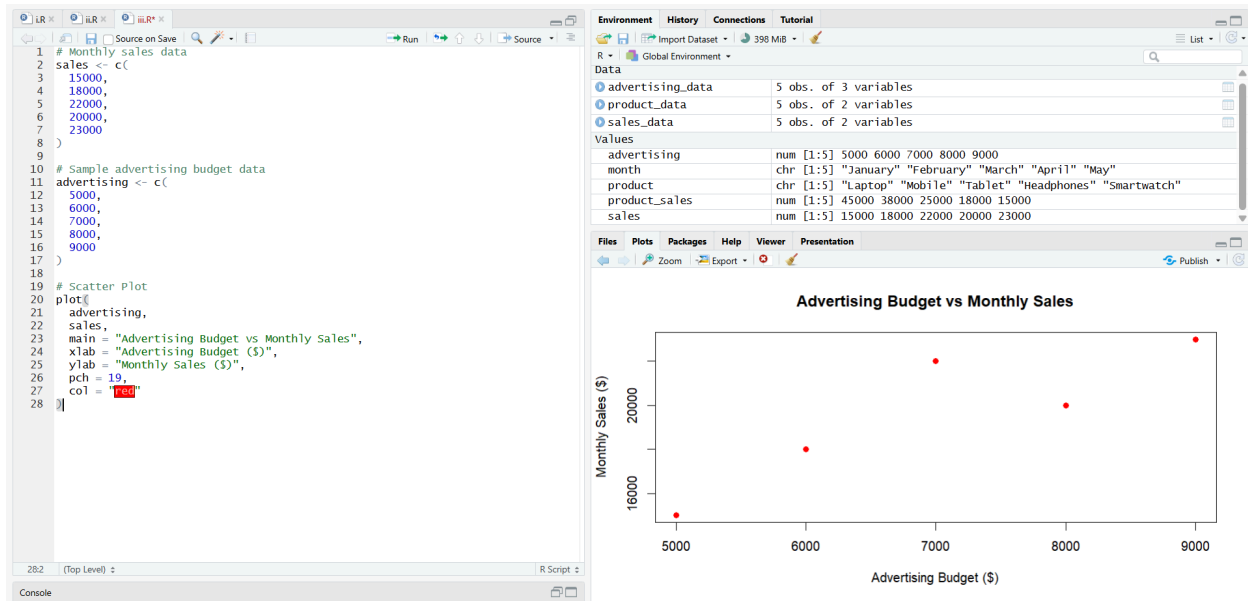
```

      ylab = "Monthly Sales ($)",
      pch = 19,
      col = "red"
    )

```

Output

Advertising Budget vs Monthly Sales Scatter Plot



Insight

The scatter plot shows a generally positive relationship between advertising budget and monthly sales. As the advertising budget increases, monthly sales generally tend to increase. This indicates that higher advertising spending is associated with higher sales for the sample values used.

PROGRAM 4: Interactive Sales Dashboard

Question

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

R Program

```
# Load Shiny
library(shiny)

# -----
# Monthly Sales Data
# -----

month <- c(
  "January",
  "February",
  "March",
  "April",
  "May"
)

sales <- c(
  15000,
  18000,
  22000,
  20000,
  23000
)

# -----
# Product Sales Data
# -----

product <- c(
  "Laptop",
  "Mobile",
  "Tablet",
  "Headphones",
  "Smartwatch"
)

product_sales <- c(
  45000,
  38000,
  25000,
```



```

    18000,
    15000
  )
# -----
# User Interface
# -----
ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "chart",
        "Select Chart:",
        choices = c(
          "Line Chart",
          "Bar Chart"
        )
      )
    )
  ),

  mainPanel(
    plotOutput("salesPlot")

  )
)

# -----
# Server
# -----

server <- function(input, output) {

  output$salesPlot <- renderPlot({

    if (input$chart == "Line Chart") {

      # Line Chart
      plot(
        sales,

```

```

    type = "o",
    xaxt = "n",
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales",
    col = "blue"
  )

  axis(
    1,
    at = 1:5,
    labels = month
  )

} else {

  # Bar Chart
  barplot(
    product_sales,
    names.arg = product,
    xlab = "Product",
    ylab = "Annual Sales ($)",
    main = "Top-Selling Products",
    col = "skyblue"
  )

}

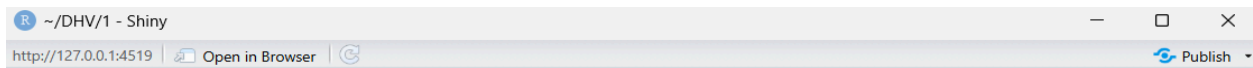
})
}

# -----
# Run Dashboard
# -----

shinyApp(
  ui = ui,
  server = server
)

```

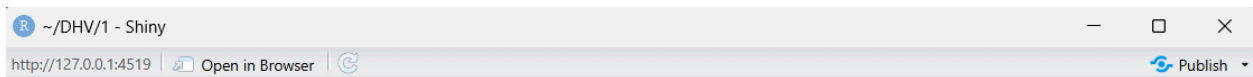
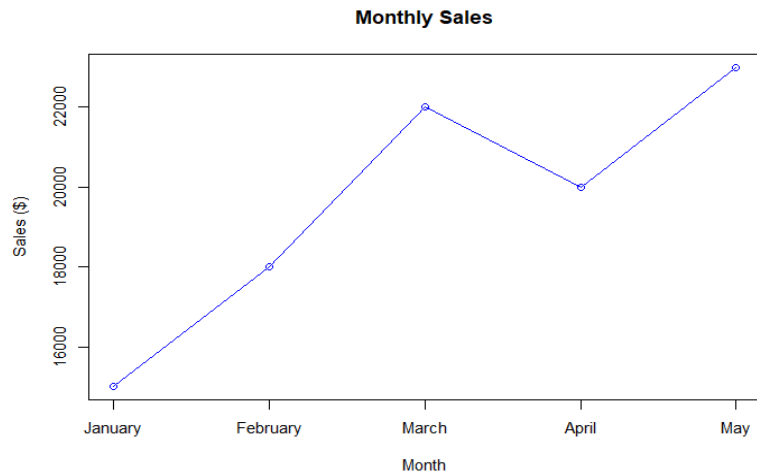
Dashboard Output



Monthly Sales Dashboard

Select Chart:

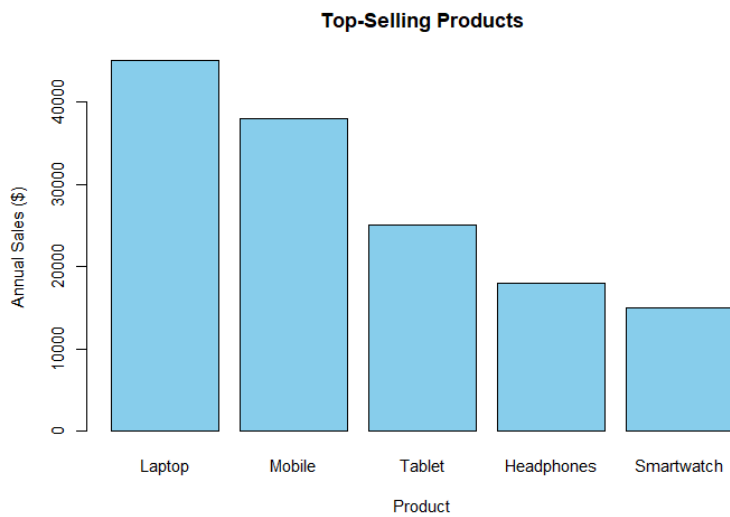
Line Chart



Monthly Sales Dashboard

Select Chart:

Bar Chart



Observation

The interactive dashboard provides a chart-selection control that allows users to switch between the monthly sales line chart and the top-selling products bar chart.

Procedure

1. Create the monthly sales vectors in R.
2. Use `plot()` to create the monthly sales line chart.
3. Add month names to the x-axis using `axis()`.
4. Create the product and annual sales vectors for the bar chart.
5. Use `barplot()` to visualize the top-selling products.
6. Create advertising-budget and monthly-sales vectors of equal length.
7. Use `plot()` to create the scatter plot.
8. Observe the relationship between advertising budget and sales.
9. Install and load the `shiny` package.
10. Create the Shiny UI and server components.
11. Run the Shiny application.
12. Use the chart-selection box to switch between the line chart and bar chart.

Result

The monthly sales data was successfully visualized using a line chart, the top-selling products were represented using a bar chart, and the relationship between advertising budget and monthly sales was explored using a scatter plot. An interactive Shiny dashboard was also created to allow users to switch between the line chart and bar chart.